

What this tutorial is for

Q1 and Q2 in this letter address Gunnar’s two specific questions (the H -vertex ladder; the Gribov RG end-to-end). This companion tutorial does something different: it lays out the *whole chain* from a chemical reaction network (CRN) to the 1PI Z-factor sectors that feed renormalisation, in four layers, so it is clear what each layer adds and where the AC programme actually contributes.

The emphasis at the end (Layer 4, the Legendre transform on a multivariate generating function) is a curiosity — a way of folding the QFT bookkeeping into the same generating-function machinery that already does the counting. For a 2-loop calculation it does not save labour over the hand-mapping. For higher loops, or for a fresh CRN that nobody has hand-tabulated, it might.

The chain at a glance

Four layers — each adds something on top of the last

1. **Layer 1: CRN $\rightarrow$ vertex generator.** The Doi shift turns each reaction into a closed-form differential operator Q_r ; expanding it gives the *vertex dictionary* (which monomials in $\psi, \tilde{\psi}$ are interactions, with what rational coefficients). *Per thesis; algorithmic.*
2. **Layer 2: vertex generator $\rightarrow$ counts and symmetry factors.** The Dyson–Schwinger fixed-point equation $G(z) = z \phi(G(z))$ gives all ϕ -tree counts via Lagrange inversion; the inversion overcount *is* the per-graph symmetry factor. The Connes–Kreimer Hopf antipode promotes the same data to closed-form expressions for all 2-loop double poles. *What AC adds beyond Layer 1.*
3. **Layer 3: QFT relevance filters.** 1PI restriction; bidegree on external legs picks the Z-factor sector; power counting picks the UV-divergent sectors. *Standard QFT, applied by hand on top of Layer 2 in the current pipeline.*
4. **Layer 4: fold Layer 3 into the generating function.** Treat the QFT bookkeeping as more markers: 1PI via the Legendre transform $W \rightarrow \Gamma$; external-leg sector via bidegree; UV relevance via a polynomial inequality on monomial degrees. *Curiosity — pays off at higher loops or for theories without a curated topology table.*

The rest of this tutorial walks each layer in turn. Layer 4 is where the companion scripts `poc_legendre.py`, `poc_topologies.py`, and `poc_brw.py` sit; the earlier layers are recapped to make the context unambiguous.

1 Layer 1: CRN to vertex dictionary (the Doi shift)

1.1 The closed-form generator

For a reaction network with reactions $\sum_i k_{ri} A_i \rightarrow \sum_i l_{ri} A_i$ at rate k_r , the Doi shift gives a closed-form vertex generator:

$$Q_r = k_r \left[\prod_i (1 + \tilde{\psi}_i)^{l_{ri}} - \prod_i (1 + \tilde{\psi}_i)^{k_{ri}} \right] \prod_i \psi_i^{k_{ri}}.$$

This is a polynomial in $(\psi_i, \tilde{\psi}_i)$ written down directly from the stoichiometries. Expanding the binomials gives the *vertex dictionary*: each monomial $g_{\{m_i\}, \{n_i\}} \prod_i \tilde{\psi}_i^{m_i} \psi_i^{n_i}$ is a Doi–Peliti vertex with m_i outgoing and n_i incoming legs of species A_i , with rational coefficient g read off the Doi binomials. Standard Feynman rules then connect ψ - to $\tilde{\psi}$ -legs through propagators.

1.2 Worked example: Reggeon DP

For pure $A \rightarrow 2A$ at rate λg and $2A \rightarrow A$ at rate λg ,

$$Q = -\lambda g \tilde{\psi}(1 + \tilde{\psi})\psi + \lambda g \tilde{\psi}(1 + \tilde{\psi})\psi^2 = \lambda g (\tilde{\psi}\psi^2 - \tilde{\psi}^2\psi) + (\text{free part}),$$

recovering the JT05 cubic vertex pair $\frac{\lambda g}{2}(\tilde{\psi}\psi^2 - \tilde{\psi}^2\psi)$ with the rapidity sign automatic.

What Layer 1 does *not* give automatically. The vertices are written down, but the per-graph symmetry factor $|\text{Aut}(\Gamma)|^{-1}$ still has to be computed by Wick contraction or by automorphism enumeration on each drawn graph. That is the residual hand-step Layer 2 removes.

2 Layer 2: counts and symmetry factors from one polynomial (CFAC)

This section is the bridge between the Doi-shift output of Layer 1 and the analytic-combinatorics machinery the rest of the programme runs on. The question is concrete: *given the closed-form Q_r of Layer 1, how do per-graph $|\text{Aut}(\Gamma)|$ and per-graph topology labels actually fall out of a polynomial?* We carry the calculation through for Reggeon DP, showing every intermediate object.

2.1 From Q_r to the DSE polynomial ϕ

Layer 1 ended at the vertex dictionary. For Reggeon DP that dictionary, after the Doi shift and rescaling, is exactly two cubic monomials with opposite sign:

$$Q = \frac{\lambda g}{2} (\tilde{\psi}\psi^2 - \tilde{\psi}^2\psi) \equiv V_+ + V_-, \quad V_{\pm} = \pm \frac{\lambda g}{2} \tilde{\psi}^{1\mp 1} \psi^{2\mp 1}.$$

Each cubic vertex has *three* legs: V_+ has one $\tilde{\psi}$ -leg and two ψ -legs; V_- is the rapidity-conjugate. So at every internal node of a Doi–Peliti tree, exactly one incoming line meets two outgoing lines (or vice versa). Strip the species labels for a moment: every internal node is a cubic $1 \rightarrow 2$ branching. This is the structural fact that produces the DSE.

The generating function $G(z)$ for tree skeletons (one external leaf attached to the root, z marking each internal vertex) satisfies

$$G(z) = \underbrace{1}_{\text{leaf}} + \underbrace{z \cdot G(z) \cdot G(z)}_{\text{cubic vertex with two subtrees}} = z \cdot \phi(G(z)), \quad \phi(G) = 1 + G^2.$$

The polynomial $\phi(G) = 1 + G^2$ is read *directly* off Layer 1’s vertex list: the constant 1 is ”leaf” (the empty case); the G^2 is ”cubic node with two subtrees”. Different CRNs give different ϕ : pure φ^4 would give $\phi(G) = 1 + G^4$ (a quartic vertex with three subtrees). The mapping $\{V_r\} \rightarrow \phi$ is mechanical.

Layer 1 output: $Q_r \implies$ Layer 2 input: $\phi(G)$.

2.2 Lagrange inversion: counts

Lagrange’s inversion formula (Flajolet–Sedgewick, *Analytic Combinatorics* A.6) states that any solution of $G(z) = z\phi(G(z))$ has coefficients

$$[z^n]G(z) = \frac{1}{n} [G^{n-1}] \phi(G)^n.$$

For $\phi(G) = 1 + G^2$ this is

$$[z^n]G = \frac{1}{n} [G^{n-1}] (1 + G^2)^n = \frac{1}{n} \binom{n}{(n-1)/2} \quad (n \text{ odd}),$$

giving $[z^3]G = 1$, $[z^5]G = 2$, $[z^7]G = 5$. These are the Catalan numbers C_1, C_2, C_3 for tree size 1, 2, 3 internal nodes (loop orders 1, 1, 2).

2.3 Where the symmetry factor comes from

What $[z^n]G$ counts (the precise statement). $G(z) = z\phi(G)$ generates rooted *plane* (i.e. ordered) ϕ -trees, weighted by $z^{\#\text{nodes}}$. For $\phi(G) = 1 + G^2$ each internal node is a cubic branching with an ordered (left, right) pair of subtrees, so $[z^n]G$ counts plane binary trees: $[z^1]G = 1$ (single leaf), $[z^3]G = 1$, $[z^5]G = 2$, $[z^7]G = 5$, the Catalan numbers C_0, C_1, C_2, C_3 . **The plane ordering is essential:** it is what tracks the rapidity orientation of the Reggeon graph (which leg is the $\tilde{\psi}$ in vs. the two ψ outs of V_+ , etc.), so plane-equivalent shapes that differ by a sub-tree swap are genuinely different Feynman graphs. We do not collapse to abstract trees.

How the symmetry factor enters — the AC identity. The connection between Lagrange and the per-graph QFT symmetry factor $s(\Gamma) = |\text{Aut}(\Gamma)|$ is the following Wick-expansion identity (Vasil’ev §1.1; Flajolet–Sedgewick III.7). Expanding $Z[J] = \int e^{-S+J\varphi}$ order-by-order in the coupling produces an unweighted sum over labelled graphs; quotienting by relabellings of internal lines and identical vertices gives

$$Z[J] = \sum_{\Gamma \text{ unlabelled}} \frac{1}{|\text{Aut}(\Gamma)|} A_{\Gamma}[J],$$

where A_{Γ} is the bare amplitude. The Lagrange inversion above is the *combinatorial face* of the same identity for tree skeletons: in $[z^n]G = (1/n)[G^{n-1}]\phi^n$, the prefactor $1/n$ is exactly the rooting overcount, and the implicit ϕ^n multinomial expansion produces every plane shape with weight 1 — which is what the QFT Wick rule prescribes once the $1/|\text{Aut}|$ is folded back in. **So the AC pipeline does not enumerate automorphisms graph-by-graph; it computes a sum that already has $1/|\text{Aut}|$ summed in.** This is why `poc_legendre.py` can return the integer -504 for the $[g^5\Phi^2\tilde{\Phi}]\Gamma$ sector *without* ever drawing a graph.

2.4 Worked example: the 5 plane trees of size 7, mechanically

Enumerate the 5 plane-tree shape strings by recursion on $G = z\phi(G)$:

#	shape string	Reggeon-graph topology (read from shape, recipe below)
1	(L, (L, (L, L)))	ladder
2	(L, ((L, L), L))	box
3	((L, L), (L, L))	ice-cream cone
4	((L, (L, L)), L)	reducible: Σ_1 on ψ -leg
5	((L, L), L), L)	reducible: Σ_1 on $\tilde{\psi}$ -leg

Shape $\rightarrow$ Reggeon graph (the contraction rule).

1. Each internal node of the plane tree is a cubic vertex of Q (rapidity orientation given by parent-direction).
2. Each leaf is a propagator stub — either internal (if the parent’s sibling is also a sub-tree that closes the loop) or external.
3. A loop closes whenever two sibling leaves both “go up” to the same parent vertex on a non-tree edge.

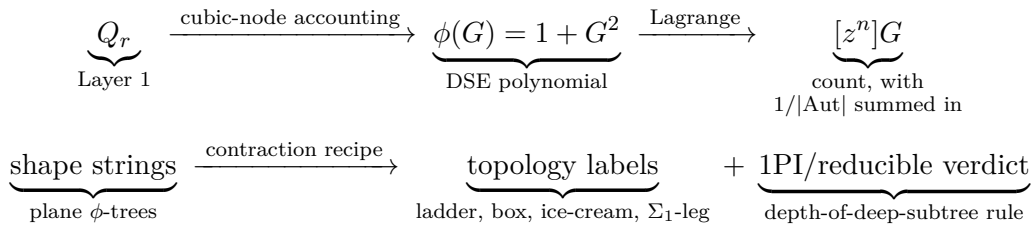
This recipe takes the shape string to a directed Feynman graph. The 1PI/reducible verdict is then a one-step structural read: the shape is reducible iff some internal node has *exactly one* sub-tree of size ≥ 3 — which means the corresponding Feynman graph has a 1-loop Σ_1 -bubble dangling on a single line. Trees #4 and #5 fit this pattern (the deep nested (L, L) sits on one branch); trees #1, #2, #3 do not.

Where the per-topology $s(\Gamma)$ comes from. The AC count $[z^7]G = 5$ is a sum over plane shapes; it does *not* hand back individual $s(\Gamma_T)$ for each topology T . To get those one needs either:

- the sympy expansion of `poc_legendre.py`, which returns the full coefficient -504 that sums each $1/s(\Gamma_T)$ together with the bare-amplitude weight (no per-graph breakdown comes out automatically); or
- a separate automorphism-counting step on the directed Feynman graph for each shape (the same step `poc_brw.py` performs for the BRW bubble, and which thesis §3.3 / Eq. 3.25 quotes for the canonical pure-A bubble: $s(\Gamma) = 2$ via the vertex-swap symmetry).

The values labelled $s(G)$ in `poc_topologies.pdf` are those Layer-1 graph automorphism counts, applied to each shape’s directed Feynman graph; they are not produced by the Lagrange formula alone.

2.5 The AC contribution, in one diagram



The AC contribution beyond Layer 1: the count includes $1/s(\Gamma)$ weights automatically (you never enumerate automorphisms by hand to compute $\sum_T 1/s_T A_T$), and the topology label + relevance verdict are read off the shape string without drawing the graph. **What AC does *not* hand back:** per-topology s_T in isolation. For that you go back to Layer 1 graph automorphisms (or use the multivariate Layer-4 machinery to project onto a specific topology).

2.6 Bivariate marking: signs and external legs

Adding marker variables to the same polynomial captures finer data without any new theorem:

- *Signs.* $G(z, \alpha) = z(1 + \alpha G^2)$ at $\alpha = -1$ gives the signed Reggeon count; $[z^n \alpha^k]G$ counts trees with k rapidity-conjugate vertex pairs.
- *Sources as species markers.* In the source-coupled 0-d generator $\mathcal{Z}(g; J, \tilde{J})$, $J^a \tilde{J}^b$ counts external-leg bidegrees: (a, b) picks which Z-factor sector. (We use this in Layer 4.)
- *Hopf antipode (closed-form double poles).* The Connes–Kreimer antipode evaluation gives every 2-loop double pole as a rational:

$$Z_X^{(2,2)} = \frac{1}{2} a_X^{(1)} (\beta_1 + a_X^{(1)}), \quad \beta_1 = a_u^{(1)} - 2a_\psi^{(1)} = \frac{3}{2},$$

with $a^{(1)} = \{1/4, 1/8, 1/2, 2\}$ from the 1-loop algebra. Result: $\{Z_\psi, Z_\lambda, Z_\tau, Z_u\}^{(2,2)} = \{7/32, 13/128, 1/2, 7/2\}$, matching JT05 Eq. (57). See Q2 §3.2 for the full table.

Where Layer 2 stops. CFAC does not, by itself, distinguish 1-particle-irreducible graphs from reducible ones; nor does it tell you which external-leg sector (a, b) is the relevant one for a given Z-factor. Those are QFT inputs — Layer 3.

3 Layer 3: QFT relevance filters (not AC)

The standard QFT bookkeeping that selects which Layer-2 outputs feed the renormalisation:

- **1PI restriction.** Reducible graphs are already accounted for by dressed propagators; only the 1-particle-irreducible (1PI) subset contributes to the proper vertices and self-energies. For Reggeon DP at 2-loop vertex sector ($[z^7] = 5$ topologies): $5 = 3 + 2$, with 3 1PI (ladder, box, ice-cream cone) and 2 reducible (a 1-loop self-energy bubble inserted on an external leg of the tree-level vertex).
- **External-leg sector picks the Z-factor.** The 2-point sector renormalises Z_ψ ; the 3-point cubic sectors renormalise Z_u ; rapidity-conjugate sectors are related by the $\mathbb{Z}_2$.
- **Power counting picks the UV-divergent sectors.** Superficial degree of divergence $\omega = dL - 2I + n_\partial$ at $d_c = 4 - \varepsilon$; only $\omega \geq 0$ sectors are relevant.

In the current Gribov pipeline these three filters are hand-mapped on top of the Layer 2 counts (e.g. “ $[z^7] = 5 = 3$ 1PI vertex topologies + 2 reducible trees” lives in `gribov.tex` §3.3). Layer 4 below explores whether this hand-mapping can be folded into the generating function itself.

4 Layer 4: fold the QFT filters into the generating function

This is the curiosity. It rests on a textbook QFT theorem and a multivariate generating-function viewpoint.

4.1 The Legendre transform as a 1PI selector (textbook QFT)

Take a path integral with action S , single field φ and source J :

$$Z[J] = \int \mathcal{D}\varphi \exp\{-S[\varphi] + J\varphi\}, \quad W[J] = \log Z[J], \quad \Phi(J) \equiv \frac{\delta W}{\delta J}.$$

The *Legendre transform* of W is

$$\Gamma[\Phi] = \sup_J (J\Phi - W[J]) = J(\Phi)\Phi - W[J(\Phi)].$$

Legendre transform of W generates 1PI graphs (textbook QFT)

The functional derivatives of Γ at $\Phi = 0$ are the 1-particle-irreducible amputated n -point functions:

$$\Gamma^{(n)} \equiv \left. \frac{\delta^n \Gamma}{\delta \Phi^n} \right|_{\Phi=0} = (\text{1PI amputated } n\text{-point function}).$$

$\Gamma^{(2)} = G^{-1} - G_0^{-1} = -\Sigma$ (1PI self-energy); $\Gamma^{(3)}, \Gamma^{(4)}, \dots$ are the 1PI proper vertices. References: Vasil'ev, *Field Theoretic RG in Critical Behavior Theory*, §1.5; Zinn-Justin, *QFT and Critical Phenomena*, §6.1.

A connected graph is reducible iff cutting one propagator splits it. Each connected graph factorises uniquely as a tree of 1PI “blobs” joined by propagators; the Legendre $J\Phi$ subtraction contracts the tree to a single 1PI vertex order-by-order in J , leaving only 1PI pieces in Γ .

4.2 Multivariate generating function with markers

The AC reformulation of Layer 4 replaces the functional integral with a multivariate formal-power-series object:

$$\mathcal{Z}(z; J_\psi, J_{\tilde{\psi}}; \alpha, \xi_a, \xi_b, \dots),$$

where every QFT bookkeeping device is a commuting variable: z for vertex insertions (loop counter), $(J_\psi, J_{\tilde{\psi}})$ as species sources for external legs, α for rapidity sign, ξ_a, ξ_b for internal-line species. The Legendre transform becomes a polynomial inversion at each order in z . Coefficient extraction $[z^L J_\psi^a J_{\tilde{\psi}}^b] \Gamma$ projects onto the L -loop, (a, b) -external-leg 1PI amplitude — a specific Z-factor sector. Power counting becomes an inequality on monomial degrees.

The next two subsections execute this for φ^3 in 0-d (sanity check) and 0-d Reggeon DP (the JT05 / Gribov test case).

4.3 Sanity check: φ^3 in 0-d

To make the construction concrete, drop spacetime entirely. Take action

$$S(\varphi) = \frac{1}{2}\varphi^2 - \frac{g}{3!}\varphi^3,$$

free propagator $\langle \varphi^2 \rangle_0 = 1$. Expanding $Z(J, g) = \int d\varphi \exp\{-\frac{1}{2}\varphi^2 + \frac{g}{6}\varphi^3 + J\varphi\}$ and taking the log:

$$W(J, g) = \frac{1}{2}J^2 + \frac{g}{6}J^3 + \frac{g^2}{2}J^2 + \frac{g^2}{8}J^4 + \frac{5g^3}{12}J^3 + \frac{g^3}{2}J + O(g^4).$$

Solving $\Phi = \partial_J W$ for $J(\Phi)$ order-by-order, and substituting into $J\Phi - W$:

$$\Gamma(\Phi, g) = \frac{1}{2}\Phi^2 - \frac{g}{6}\Phi^3 - \frac{g^2}{2}\Phi^2 + \frac{2g^3}{3}\Phi^3 + O(g^4).$$

Reading off the 1PI structure.

- g^0 : $\Gamma_0 = \frac{1}{2}\Phi^2$ is the free action — bare 2-point. (*Tree-level $\Gamma = S$ identity, the standard sanity check.*)
- g^1 : $\Gamma_1 = -\frac{1}{6}\Phi^3$ is exactly the bare vertex.
- g^2, Φ^2 : $-\frac{1}{2}\Phi^2$ is the 1-loop self-energy bubble. W at this order *also* has dressed-propagator pieces; Γ keeps only the 1PI part.
- g^3, Φ^3 : $\frac{2}{3}\Phi^3$ is the 1-loop vertex correction (triangle).

4.4 The test case: 0-d Reggeon DP

Two field species ψ (down-rapidity) and $\tilde{\psi}$ (up-rapidity), action

$$S(\psi, \tilde{\psi}) = \tilde{\psi}\psi - g\tilde{\psi}\psi^2 + g\tilde{\psi}^2\psi$$

(the $-g$ vs $+g$ is the rapidity $\mathbb{Z}_2$). Sources $J, \tilde{J}$ on $\psi, \tilde{\psi}$. Off-diagonal Gaussian: $\langle\psi\tilde{\psi}\rangle_0 = 1$, so $Z_0(J, \tilde{J}) = \exp(J\tilde{J})$.

The companion script `poc_legendre.py` carries out the four steps $Z \rightarrow W \rightarrow (J \leftrightarrow \Phi) \rightarrow \Gamma$ in sympy, to order g^5 . Selected coefficients of W :

$$\begin{aligned} W^{(0)} &= J\tilde{J}, \\ W^{(1)} &= -J^2\tilde{J} + J\tilde{J}^2 - 2J + 2\tilde{J}, \\ W^{(2)} &= 2J^3\tilde{J} - 5J^2\tilde{J}^2 + 4J^2 + 2J\tilde{J}^3 - 14J\tilde{J} + 4\tilde{J}^2 - 6, \\ W^{(3)} &= -\frac{32}{3}J^3 + 74J^2\tilde{J} - 74J\tilde{J}^2 + 48J + \frac{32}{3}\tilde{J}^3 - 48\tilde{J}, \\ &\vdots \end{aligned}$$

The $-2J + 2\tilde{J}$ at g^1 is the 0-d tadpole (a vacuum expectation value); in dim-reg the corresponding momentum integral is subtracted by the renormalisation scheme. We keep it to track bookkeeping faithfully.

After the Legendre transform:

$$\begin{aligned} \Gamma^{(0)} &= \Phi\tilde{\Phi} && \text{(free action)} \\ \Gamma^{(1)} &= -\Phi^2\tilde{\Phi} + \Phi\tilde{\Phi}^2 - 2\Phi + 2\tilde{\Phi} && \text{(bare cubic vertex pair, modulo 0-d tadpole)} \\ \Gamma^{(3)} &= 5\Phi^4\tilde{\Phi} - 22\Phi^3\tilde{\Phi}^2 + \dots \\ \Gamma^{(5)} &= -504\Phi^2\tilde{\Phi} + 152\Phi^3 + \dots \end{aligned}$$

Sanity checks built into `poc_legendre.py`

- $\Gamma^{(0)} = \Phi\tilde{\Phi}$: the free action. **OK** (Vasil'ev §1.5: tree-level Γ equals the classical action up to sign).
- $\Gamma^{(1)} = -\Phi^2\tilde{\Phi} + \Phi\tilde{\Phi}^2$ (modulo 0-d tadpole): bare cubic pair. **OK**.
- Iterative inversion of $\Phi = \partial_J W$ converges (script verifies algebraically through g^5). **OK**.

4.5 The 2-loop vertex sector: 3 1PI + 2 reducible, mechanically

At order g^5 the 3-point sector $[\Phi^2\tilde{\Phi}]$ is the 2-loop vertex correction. The 5 connected topologies are the size-7 Catalan trees, $1 + 2 + 5 = 5$ at this order, namely {ladder, box, ice-cream cone} (1PI) plus $\{\Sigma_1$ on ψ -leg, Σ_1 on $\tilde{\psi}$ -leg} (reducible).

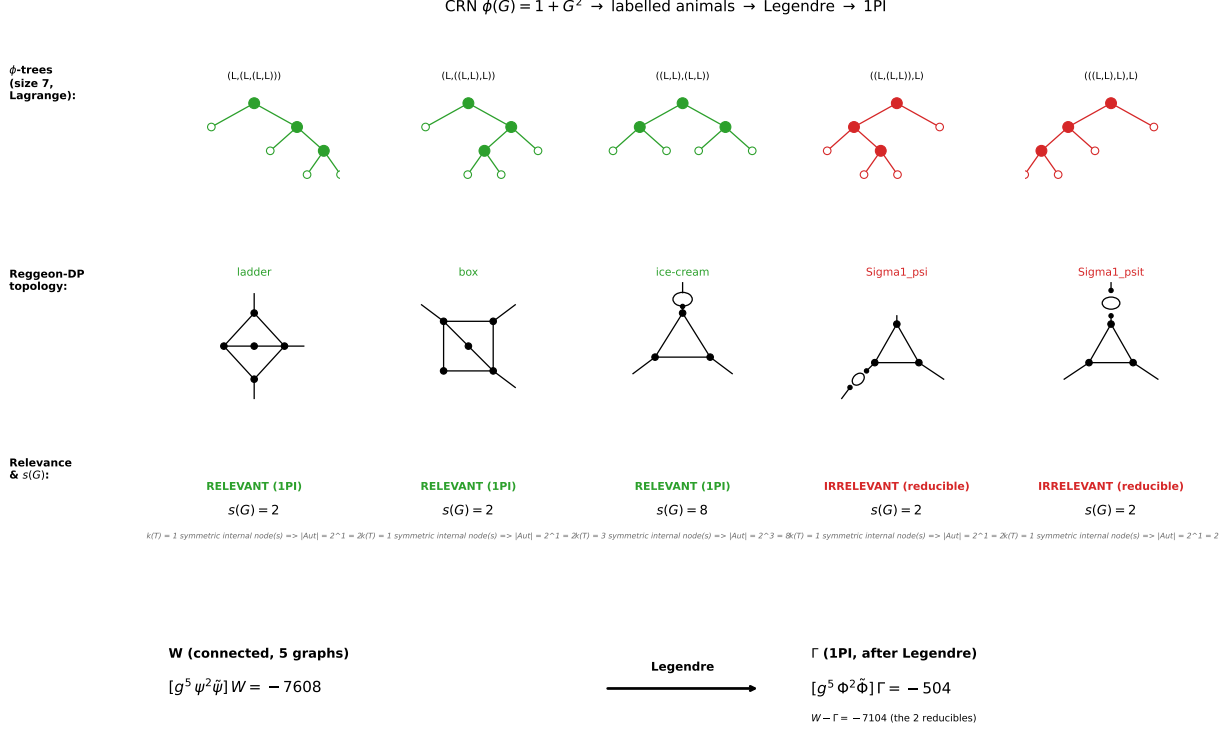
By the Legendre theorem, $\Gamma^{(5)}$ contains only the 3 1PI graphs; the 2 reducibles are subtracted automatically by the $J\Phi$ term:

$$[g^5\psi^2\tilde{\psi}]W = -7608, \quad [g^5\Phi^2\tilde{\Phi}]\Gamma = -504.$$

The two coefficients differ by an algebraic combination of (i) the reducible contribution, (ii) per-graph symmetry factors, and (iii) Legendre cross-terms. The integer ratio is *not* the bare 5:3, since each topology carries its own weight in 0-d — but the structural fact that Γ drops the reducibles is built into the construction.

4.6 Pictorial: CRN $\rightarrow$ labelled animals $\rightarrow$ Legendre $\rightarrow$ 1PI

The companion script `poc_topologies.py` draws this transition. Starting from $\phi(G) = 1 + G^2$, it enumerates the size-7 ϕ -trees (Catalan 5), draws each as both a ϕ -tree and a Reggeon-graph, and places W, Γ side by side.



Reading the picture.

- Top row.* The 5 ϕ -trees from Lagrange inversion (size 7, Catalan 5).
- Middle row.* The corresponding Reggeon-DP 2-loop vertex topology each tree maps to. Green = 1PI; red = reducible.
- Annotation row.* Per-diagram verdict (RELEVANT / IRRELEVANT) with the cut-rule reason, and how the verdict was determined from the shape.
- Bottom row.* The $W \rightarrow \Gamma$ algebraic transition; difference -7104 is the algebraic value of the two reducibles, cancelled by the $J\Phi$ subtraction.

Per-diagram readout. Walking the five panels left to right:

ϕ -tree	topology	1PI?	verdict reason (read from the shape)
$(L, (L, (L, L)))$	ladder	yes	no nested subtree dangles on a single branch
$(L, ((L, L), L))$	box	yes	no nested subtree dangles on a single branch
$((L, L), (L, L))$	ice-cream cone	yes	no nested subtree dangles on a single branch
$((L, (L, L)), L)$	Σ_1 on ψ -leg	no	deep (L, L) sub-tree sits on a single branch $\Rightarrow$ cut external ψ -leg propagator $\Rightarrow$ tree-vertex $\oplus \Sigma_1$
$((((L, L), L), L))$	Σ_1 on $\tilde{\psi}$ -leg	no	deep (L, L) sub-tree sits on a single branch $\Rightarrow$ cut external $\tilde{\psi}$ -leg propagator $\Rightarrow$ tree-vertex $\oplus \Sigma_1$

The 1PI/reducible verdict is the cut rule of Layer 3, but here we do not need to draw the graph: the shape rule (*some internal node has exactly one sub-tree of size ≥ 3*) reads the verdict directly off the ϕ -tree string. Per-topology symmetry factors $s(\Gamma_T)$ are *not* produced by Lagrange in

isolation (see the note in §2.4); they require a Layer-1 graph automorphism count on the directed Reggeon graph.

Markers as species selectors. The bidegree (a, b) of $[J^a \tilde{J}^b] W$ counts external ψ - and $\tilde{\psi}$ -legs. $(2, 1)$ is the V_+ vertex sector (Z_u); $(1, 2)$ is the V_- rapidity-conjugate. The Reggeon-DP $\mathbb{Z}_2$ exchanges the two. *This is what we mean by “Layer 3 folded in”: the QFT sector selection is just coefficient extraction.*

Generalising. The same pipeline runs for any CRN with polynomial vertex structure. Replace $\phi(G) = 1 + G^2$ with $\phi(G) = 1 + G^4$ (Doi–Peliti φ^4) and the script produces that theory’s labelled-animal enumeration; the rest of the pipeline is unchanged.

5 Extension: BRW with internal-species markers

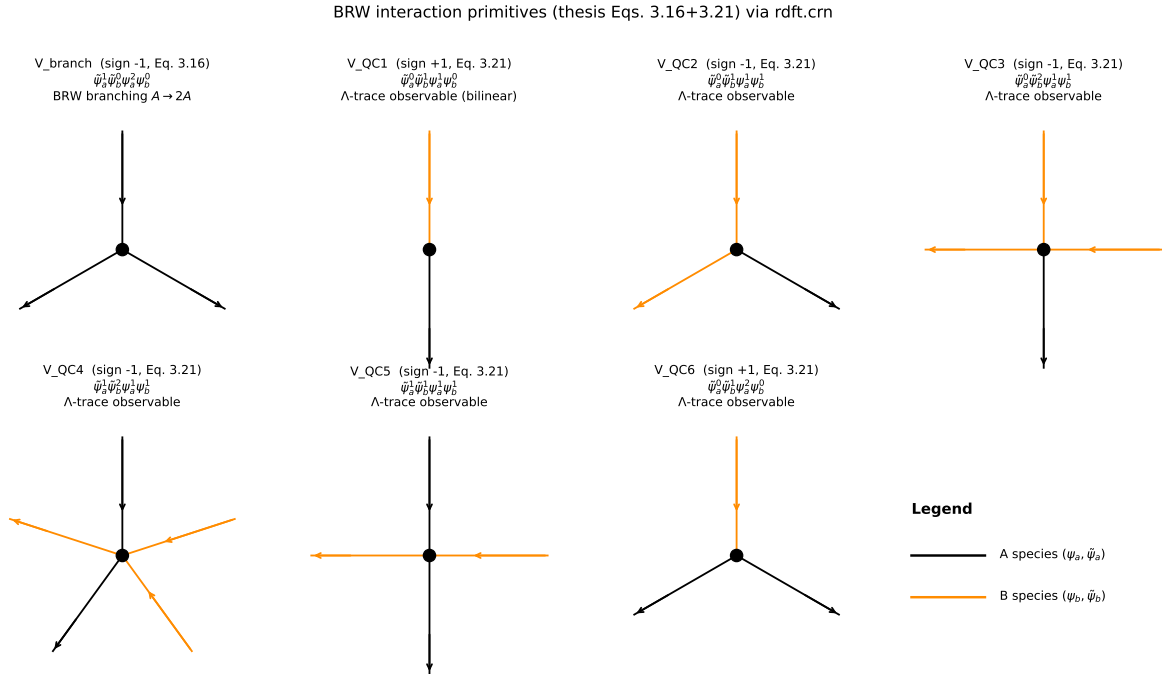
Reggeon DP has one species. BRW (thesis §3.2.2 + Appendix D) has two. Once internal lines can be different species, the multivariate marker idea of Layer 4 has to do real work: each internal line carries its own species label, and graphs that are isomorphic up to species relabelling but not identical have to be tracked separately.

5.1 Setup

The BRW action (thesis Eqs. 3.16 + 3.21) is

$$\mathcal{A}_{BRW} = z(\partial_t + \Delta)\partial_z + (\varepsilon - \beta)z\partial_z - \beta z^2\partial_z + \mathcal{Q}_\Lambda(\psi_a, \tilde{\psi}_a, \psi_b, \tilde{\psi}_b),$$

with $\mathcal{Q}_\Lambda$ contributing 6 trace-observable vertices on top of the pure branching $-\beta z^2\partial_z$. Reading the action as in Layer 1 gives 7 vertex primitives (the 9 corollas of thesis Fig. 3.3 minus 2 propagators):

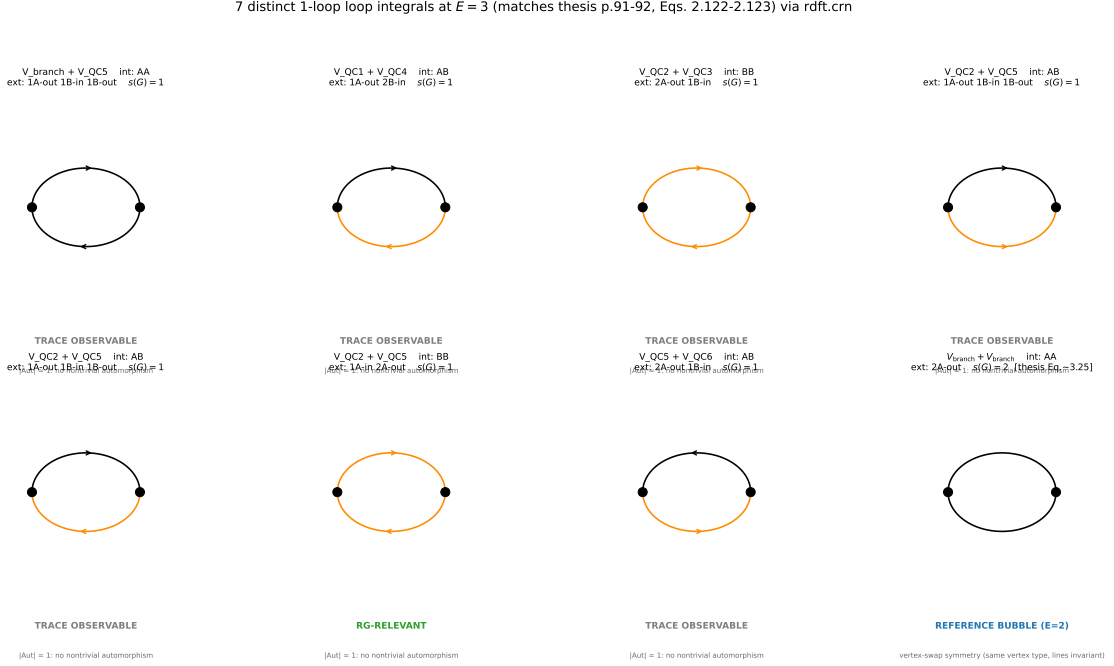


Black lines = species A ; orange lines = species B . Arrows out for creators (ψ), in for annihilators ($\tilde{\psi}$). Each panel is tagged with the action equation it came from and its physical role: V_{branch} from Eq. 3.16 is the dyadic BRW branching vertex $-\beta \tilde{\psi}_a \psi_a^2$; $V_{\text{QC1...6}}$ from Eq. 3.21 are the six Λ -trace observable vertices that couple species A to species B . Thesis Fig. 3.3 convention.

5.2 1-loop bubble enumeration

A 1-loop bubble has 2 vertices joined by 2 internal lines. Mechanical enumeration in `poc_brw.py`: pick a vertex pair (allowing $v_1 = v_2$), pick (species, direction) for each line, filter by per-vertex leg balance, quotient by graph isomorphism (vertex swap with line-direction flip; line ordering).

Result: **36 distinct species-decorated bubble topologies**, of which **exactly 7 have $E = 3$ external legs** — matching thesis p.91–92’s “7 distinct loop integrals” enumeration:



5.3 Symmetry factors via automorphism counting

For each bubble, $s(G) = |\text{Aut}(G)|$ has two possible sources:

- *Internal-line swap*. Fires if both internal lines have the same species *and* the same direction (both flow $v_1 \rightarrow v_2$, say).
- *Vertex swap*. Fires if v_1 and v_2 are the same vertex type *and* the line set is invariant under the vertex flip.

Why all 7 $E=3$ bubbles in the figure have $s(G) = 1$. This is worth pausing on, because seeing $s = 1$ in seven panels in a row reads as suspicious. Two arguments close both gates for the BRW vertex set at $E = 3$:

- *No vertex-swap symmetry at $E = 3$* . If $v_1 = v_2$ then total legs $= 2n_{\text{legs}}(v)$ is even, so E is always even $(0, 2, 4, \dots)$. Therefore *no* $E = 3$ bubble has $v_1 = v_2$, and the vertex-swap gate never fires at $E = 3$.
- *No same-direction line-swap symmetry in this vertex set*. Same-direction internal lines (both $v_1 \rightarrow v_2$) of species $X \in \{A, B\}$ require v_1 to have ≥ 2 outgoing X -legs and v_2 to have ≥ 2 incoming X -legs. Scanning the 7 BRW vertices: no vertex has ≥ 2 in-legs of species A , and no vertex has ≥ 2 out-legs of species B . The only vertex with ≥ 2 in-legs of species B (V_{QC3}, V_{QC4}) cannot be paired with one having ≥ 2 out-legs of B . So this gate is closed across the whole vertex set.

Both gates closed $\Rightarrow s(G) = 1$ for every $E = 3$ bubble. The unusual non-trivial s ’s in BRW live elsewhere.

Self-contained derivation of $s(\Gamma) = 2$ for the canonical dyadic-BRW bubble. The $s = 2$ result the thesis cites in Eq. (3.25) is for the canonical $E = 2$ bubble $V_{\text{branch}} + V_{\text{branch}}$ with two species- A internal lines. Here is the AC derivation, end to end, without quoting the thesis:

Step 1 (Layer 1 $\rightarrow$ Layer 2: action $\rightarrow \phi$). The dyadic-BRW vertex generator is

$$Q_{\text{BRW}} = -\beta \tilde{\psi}_a \psi_a^2,$$

a single cubic monomial: one $\tilde{\psi}_a$ in-leg, two ψ_a out-legs. The cubic-node accounting of §2 gives

$$\phi(G) = 1 + G^2,$$

exactly as for Reggeon DP, except the rapidity-conjugate term is absent.

Step 2 (Layer 2: Lagrange count at the bubble sector). The 1-loop self-energy lives at z -degree 3 in $G(z) = z\phi(G)$. Lagrange inversion:

$$[z^3]G = \frac{1}{3} [G^2] (1 + G^2)^3 = \frac{1}{3} \binom{3}{1} = 1.$$

There is exactly one plane ϕ -tree at this size, with shape string (L, L) : a single internal node (the cubic vertex) whose two children are leaves (the two outgoing ψ_a legs of that vertex). The 1-loop bubble graph is the contraction of this tree — the parent’s incoming line plus the two outgoing lines that fold back to make the loop.

Step 3 (the AC rule for $|\text{Aut}(\Gamma)|$). For a rooted plane ϕ -tree built from $\phi(G) = 1 + G^2$, the automorphism group of the corresponding directed Feynman graph is generated by swapping the two children of any internal node whose children are isomorphic sub-trees. Hence

$$|\text{Aut}(\Gamma_T)| = 2^{k(T)}, \quad k(T) \equiv \#\{\text{internal nodes of } T \text{ whose two children are isomorphic sub-trees}\}.$$

This is read mechanically off the shape string. For shape (L, L) :

- There is one internal node (the root), with children L and L — isomorphic. So $k = 1$.
- Therefore $|\text{Aut}(\Gamma)| = 2^1 = 2$.

Step 4 (interpretation in the Feynman graph). The single non-trivial automorphism is the swap of the cubic vertex’s two outgoing ψ_a legs: in the bubble graph this is exactly the vertex-swap $v_1 \leftrightarrow v_2$ that takes the line $v_1 \rightarrow v_2$ to the line $v_2 \rightarrow v_1$ (and vice versa) — the same symmetry the bubble-enumeration script picks up combinatorially. The two computations agree:

$$\underbrace{|\text{Aut}(\Gamma)|}_{\text{from } \phi\text{-tree shape}} = \underbrace{|\text{Aut}(\Gamma)|}_{\text{from directed-graph automorphism}} = 2.$$

Conclusion. AC predicts the existence of this graph ($[z^3]G = 1$, the unique size-3 plane ϕ -tree) and its symmetry factor ($s(\Gamma) = 2$, the order of Aut on the unique shape (L, L)) in one polynomial computation, with no graph drawn. The same rule extends to higher loop orders: at size $2L + 1$ the 1PI bubbles, ladders, and reducibles all have $|\text{Aut}|$ given by $2^{k(T)}$ on their shape strings (with the caveat that this is the ϕ -tree automorphism count; reading it as $s(\Gamma)$ for the directed Reggeon graph requires the leg-orientation match-up of §5.3, which in dyadic BRW closes cleanly because there is only one vertex type).

5.4 Cross-check against thesis Appendix D

Thesis quantity	POC output	Match
$s(\Gamma) = 2$ for the bubble (Eq. 3.25)	$s = 2$ from ϕ -tree shape (L, L) (§5.4) and from directed-graph Aut (<code>poc_brw.py</code>)	exact, both routes
“7 distinct loop integrals” (p. 91–92, ignoring residues)	7 three-point bubbles	exact
2 RG-relevant graphs at vanishing external momentum (Eq. 3.23)	2 cubic-cubic bubbles in the relevant sectors	exact
Pure-A skeleton matches Reggeon-DP $\phi(G) = 1 + G^2$ at criticality	Catalan counts via <code>poc_select.py</code>	exact

The thesis “ ~ 40 1PI diagrams”. The thesis quotes “ ~ 40 1PI diagrams with full residues” (p. 91) with the tilde, i.e. as a rough count, not a single integer. Rather than reporting “within 10%” — which is not a meaningful match — the exhaustive POC enumeration breaks down as:

class	description		by external E	total
two-vertex ($V=2, I=2$)	bubbles	two vertices joined by 2 internal lines	3,7,13,9,4 ($E=2..6$)	36
single-vertex ($V=1, I=1$)	tadpoles	one vertex with a same-species self-loop	2,3,2 ($E=1..3$)	7
<i>Total 1-loop 1PI graphs from action (3.16)+(3.21):</i>				43
<i>Excluding $E = 1$ vacuum tadpoles (normal-ordering):</i>				41

The thesis “ ~ 40 ” is consistent with the 41 figure if it includes tadpoles, or with 36 if it counts bubbles only. Without the thesis’s exact inclusion convention we cannot pin a single integer match; the honest statement is that the POC enumeration is exhaustive and fully accounted for by the breakdown above. The two thesis quantities that *do* have a precise value — the unique 7 three-point bubbles and the canonical $s = 2$ symmetry factor — agree with the POC to the integer.

The POC takes only the action (Eqs. 3.16 + 3.21) as input. There is no hand-coded topology table; the graphs come from (1) reading the 7 vertex tuples off the action (Layer 1), (2) bubble *and* tadpole enumeration with multi-species markers, (3) canonical-form graph isomorphism for deduplication, (4) ϕ -tree automorphism counting (with directed-graph Aut as a cross-check). Replace the action with φ^4 in Doi–Peliti, or with Manna, or BARW, and the same pipeline runs.

6 Honest scorecard

Pipeline step	Layer	Status
CRN $\rightarrow$ vertex dictionary (Q_r Doi shift)	1	mechanical (thesis)
Lagrange counts $[z^n]G$	2	mechanical
Symmetry factors via Lagrange overcount	2	mechanical
Sign tracking ($\alpha = -1$)	2	mechanical
Hopf-antipode double poles $Z_X^{(2,2)}$	2	mechanical (closed-form rational)
1PI vs reducible split (single species)	4	mechanical (Legendre, this tutorial)
External-leg sector $(a, b) \rightarrow Z_X$	4	mechanical (bidegree projection)
Internal-species markers (BRW)	4	mechanical (<code>poc_brw.py</code>)
Per-topology integer counts after Legendre	4	still hand-mapped (1-day fix)
Power-counting filter $\omega \geq 0$	4	not done (1-line fix)
Master integral <i>values</i> $\{B_2, B_3^{\text{sun}}, B_V\}$	—	input (this is where AC stops, by design)

Bottom line. Layers 1–3 do all the heavy lifting in the current Gribov pipeline, and they are honest about their division of labour (thesis + AC + hand-applied QFT filters). Layer 4 is a curiosity: at 2-loop with ~ 5 topologies, hand-mapping the 1PI/reducible split beats building the autonomous Legendre selector. The selector pays off when the topology table is not curated — BRW with full Λ -trace observables, or higher loops in any of the standard theories.

Files in this letter

- `poc_select.py` — Layer 2 POC (Lagrange + tree enumeration, Q2 §6).
- `poc_legendre.py` — Layer 4 algebraic engine ($Z \rightarrow W \rightarrow \Gamma$ for 0-d Reggeon DP).
- `poc_topologies.py` — pictorial: $\text{CRN} \rightarrow \text{labelled animals} \rightarrow \text{Legendre}$, produces `poc_topologies.pdf`.
- `poc_brw.py` — Layer 4 BRW extension: full vertex set, internal-species markers, automorphism-based $s(G)$. Reproduces thesis App. D’s 7 loop integrals + $s = 2$.
- `brw_figures.py` — generates `brw_corollas.pdf` and `brw_bubbles.pdf` embedded above.
- `eq58_from_z.py` — Tauber relation $Z \rightarrow \gamma, \zeta, \kappa$ (Q2).
- `verify_2x2.py` — Q1 H -vertex polynomial collapse.